

CS340 Portfolio Project: Bookends Database Driven Website

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URL: <http://classwork.engr.oregonstate.edu:64413/>

Part I: Executive Summary

The Bookends initial database draft consisted of four entities: Books, Authors, Users, and Reviews. Initial feedback received on the draft had recommended implementing the ability to tag books with specific genres. The feedback was appreciated and subsequently implemented through the addition of the Genres entity.

After the completion of the Step 2 draft, the major feedback suggested implementing the On Delete cascade in the SQL file as well as changing the review_score to enum data type to restrict the values to 1-5. This feedback was then implemented for the next draft: On Delete cascade was added to the SQL file and review_score was changed to enum data type.

Step 3 feedback consisted of adding drop-down functionality for the foreign keys in the UI, adding Delete functions, as well as making the column names more human readable. All of this feedback was implemented for the next draft. Joins were added to the DMQ file. Drop-downs were also implemented for all of the choices in the UI, as well as column names changed to human readable names (i.e. User rather than users_user_id). Delete functions were added to the HTML, as they were described in the DMQ file.

Lastly, Step 4 and 5 feedback had provided insight into implementing the CRUD functions needed for the UI. A major block after Step 4 was implementing the Update function for the Books page. The feedback on solving this roadblock had included using Chrome's developer tools and ensuring that id attributes are unique across the page. This feedback was extremely helpful, and one of the major issues discovered in the code was that the id attributes were not always unique, which was causing the JavaScript to reference the wrong form's elements. Since then, a lot of issues that we have had while finishing the implementation of the CRUD functions have been solved by ensuring that the id attributes are unique. Step 5 feedback additionally provided helpful pointers on finishing the CRUD implementation needed for the remainder of the portfolio project, as well as pointing out what wasn't working correctly. Overall, the feedback provided from both the TA's and fellow students was extremely helpful in inspiring further ideas for the project, as well as helping to fix bugs and issues that weren't immediately apparent.

Part II: Project Outline

A problem found among many avid readers is the fact that it is impossible to keep track of all of the books that they have read. “Frequent readers (17% of the population) read 21 or more books in the previous 12 months” (Rainie 2024). With the number of books people read on a yearly basis, it is easy to lose track of past reads, especially when they may have been loaned from a friend, rented from the library, or read as an electronic book. Even if one were to remember all of the books that they have read, it is easy to forget your overall rating for a specific book years after reading it. Bookends database driven website is the solution to this problem.

Bookends is a database driven website that will help avid book lovers everywhere track their favorite reads. Users can search the vast collection of books and log them to their digital library. Within this library, users will be able to view their personal reviews, read dates, and ratings scored 1-5 stars for each book. Tracking recent reads will allow users to manage their book collection and stay on top of their reading goals.

Users will be able to search through thousands of books and authors, with the ability to filter through genre or ISBN. The comprehensive details of the book, genre and author entities will allow users to easily search for books to log. Users create personal accounts to track the hundreds of books that they have read, which will be stored in their personal online library. A user’s library will hold all of the books that users have read, along with their rating and/or review. Users will be able to write hundreds of reviews for each book that they have read. Users will be able to use this data, to not only track how many books they have read for the year, but reflect on their favorites through their reviews.

Part III: Database Outline

- **Books:** Records the details about the different books that a user can read and review.
 - book_id: int, NN, AI
 - title: varchar(255), NN
 - ISBN: varchar(17), NN [varchar instead of int to support dashes]
 - publication_date: date, NN

Relationships:

- Authors (M:M) facilitated by Books_has_authors, because a book can have multiple authors, and an author could have written multiple books.
 - Users (M:M) facilitated by both Users_read_books and Reviews, since a user could have read many books and a book could have been read by many users. Likewise with reviews, a user could have written many reviews and a book could have many reviews
- **Authors:** Defines the names, date of birth and brief bio of authors of books.
 - author_id: int, NN, AI
 - first_name: varchar(50), NN
 - last_name: varchar(100), NN
 - date_of_birth: date, NN
 - bio: varchar(5000)

Relationships:

- Books (M:M) facilitated by Books_has_authors.

- **Users:** Defines the username of users.
 - user_id: int, NN, AI
 - username: varchar(20), NN, Unique
 - email: varchar(255), NN, Unique
 - password: varchar(255), NN
 - reading_goal: int

Relationships:

- Users (M:M) facilitated by Users_read_books and Reviews (as noted above).

- **Reviews:** Defines the relationship between Users and Books if a user has reviewed a book
 - review_id: int, NN, AI
 - review_date: date, NN
 - review_score: ENUM, (1-5)
 - review_text: varchar(5000)
 - users_user_id: int, FK
 - books_book_id: int, NN, FK

Relationships:

- Facilitates M:M between Users and Books, since a user could have written multiple reviews, and a book may have had multiple reviews written about it.
- **Genres:** Stores information regarding the genre of a book.

- **genre_id:** int, NN, AI
- name: varchar(255), Unique
- description: varchar(5000),

Relationships:

- Has a M:M relationship with Books, facilitated by Books_has_genres, since a book can belong to multiple genres and a genre can be related to multiple books.
- **Books_has_genres:** Defines the relationship between Books and Genres.
 - books_books_id: int, NN, FK
 - genres_genre_id: int, NN, FK

Relationships:

- Facilitates M:M between Books and Genres, since a user could have read multiple books, and books may have been read by multiple users.
- **Users_read_books:** Defines the relationship between Users and Books if a user has read a book.
 - users_user_id: int, NN, FK
 - books_book_id: int, NN, FK
 - read_date: date, NN

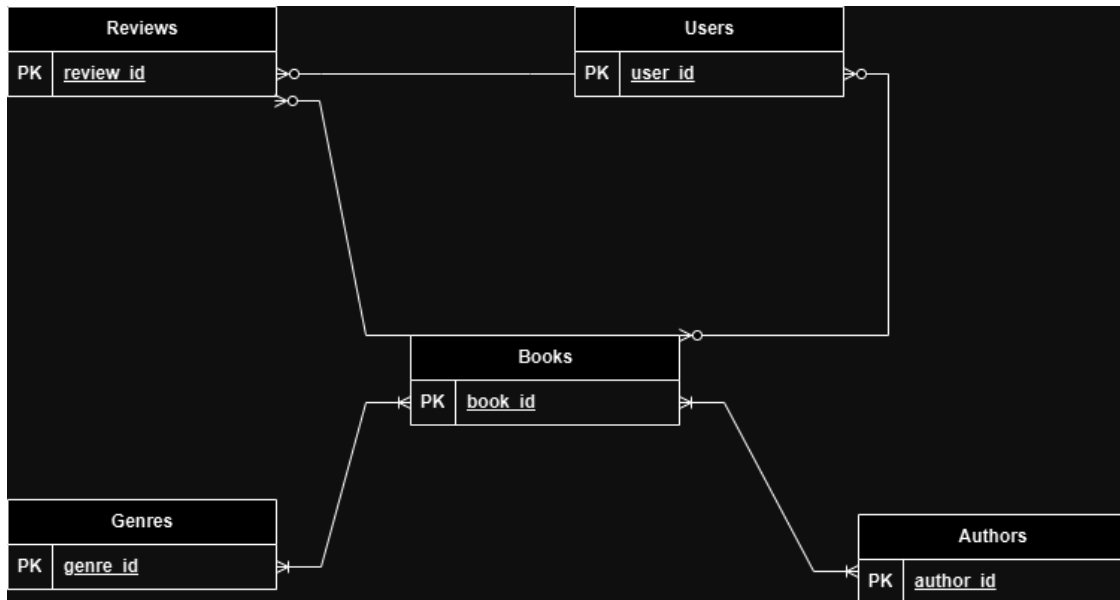
Relationships:

- Facilitates M:M between Users and Books, since a user could have read multiple books, and books may have been read by multiple users.
- **Books_has_authors:** Defines the relationship between Books and Authors
 - books_books_id: int, NN, FK
 - authors_author_id: int, NN, FK

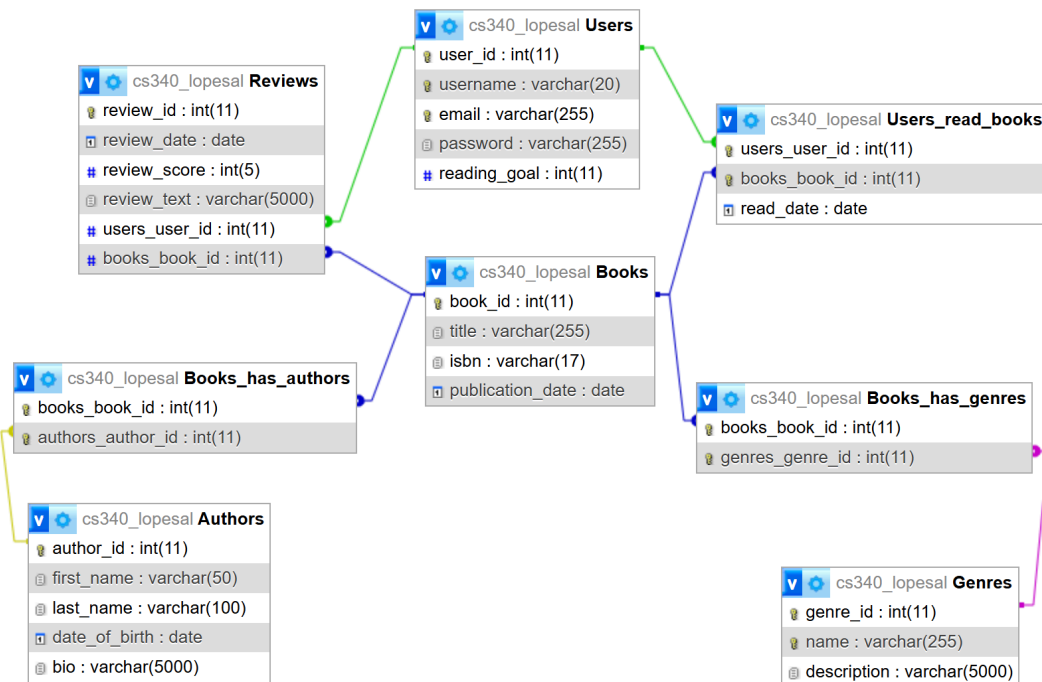
Relationships:

- Facilitates M:M between Books and Authors, since an author could have written multiple books, and books may have multiple authors.

Part IV: ER Diagram



Part V: Schema



Part VI: Sample Data

Books			
book_id	title	isbn	publication_date
1	A Tale for the Time Being	9780143124870	Tue Dec 31 2013 00:00:00 GMT-0800 (Pacific Standard Time)
2	The Hobbit	9780547928227	Tue Sep 18 2012 00:00:00 GMT-0700 (Pacific Daylight Time)
3	The Way of Kings	9780765376671	Tue Mar 04 2014 00:00:00 GMT-0800 (Pacific Standard Time)
4	Villette (Penguin Classics)	9780140434798	Tue Dec 28 2004 00:00:00 GMT-0800 (Pacific Standard Time)

Users				
user_id	username	email	password	reading_goal
1	fake_user123	fake_user@gmail.com	1234!!	30
2	second_user	Second@gmail.com	password	10
3	third_user	third@gmail.com	anotherpassword	15
4	fourth_user	fourth@gmail.com	lastpassword	45

Users_read_books		
users_user_id	books_book_id	read_date
fake_user123	A Tale for the Time Being	Wed Sep 25 2024 00:00:00 GMT-0700 (Pacific Daylight Time)
second_user	The Hobbit	Thu Oct 31 2024 00:00:00 GMT-0700 (Pacific Daylight Time)
third_user	The Way of Kings	Wed Jan 15 2020 00:00:00 GMT-0800 (Pacific Standard Time)
fourth_user	Villette (Penguin Classics)	Sat Jun 08 2019 00:00:00 GMT-0700 (Pacific Daylight Time)

Reviews

review_id	review_date	review_score	review_text	users_user_id	books_book_id
1	Wed Sep 25 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	5	Great book	fake_user123	A Tale for the Time Being
2	Thu Oct 31 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	5	Loved this one	second_user	The Hobbit
3	Wed Jan 15 2020 00:00:00 GMT-0800 (Pacific Standard Time)	5	My favorite!	third_user	The Way of Kings
4	Sat Jun 08 2019 00:00:00 GMT-0700 (Pacific Daylight Time)	5	New favorite	fourth_user	Villette (Penguin Classics)

Authors

author_id	first_name	last_name	date_of_birth	bio
1	Ruth	Ozeki	Mon Mar 12 1956 00:00:00 GMT-0800 (Pacific Daylight Time)	American-Canadian author, filmmaker and Zen Buddhist priest
2	John	Tolkein	Sun Jan 03 1892 00:00:00 GMT-0800 (Pacific Standard Time)	English writer and philologist
3	Brandon	Sanderson	Fri Dec 19 1975 00:00:00 GMT-0800 (Pacific Standard Time)	American, New York Times best-selling author of high fantasy, science fiction, and young adult books
4	Charlotte	Bronte	Sun Apr 21 1816 00:00:00 GMT-0752 (Pacific Daylight Time)	English novelist and poet, the eldest of the three Brontë sisters

Books_has_authors	
books_book_id	authors_author_id
A Tale for the Time Being	Ruth Ozeki
The Hobbit	John Tolkein
The Way of Kings	Brandon Sanderson
Villette (Penguin Classics)	Charlotte Bronte

Genres		
genre_id	name	description
1	Fantasy	Includes fantastical elements, such as magic or supernatural forces.
2	Mystery	Plot revolves around a crime or mystery to be solved.
3	Literary Fiction	Character-driven and introspective.
4	Non-Fiction	Events, people, and concepts based on fact and reality.

Books_has_genres	
books_book_id	genres_genre_id
A Tale for the Time Being	Literary Fiction
The Hobbit	Fantasy
The Way of Kings	Fantasy
Villette (Penguin Classics)	Literary Fiction

Part VII: UI Screenshots

CSS styling occurred after screenshots were taken, there is no functional change to the website

Homepage

Landing page with links to each entity page. Information on the database provided with the overview included in the homepage.

[Home](#) [Books](#) [Users](#) [Reviews](#) [Authors](#) [Genres](#)

Bookends Database

Bookends is a database driven website that will help avid book lovers everywhere track their favorite reads. Users can search the vast collection of books and log them to their digital library. Within this library, users will be able to view their personal reviews, read dates, and ratings scored 1-5 stars for each book. Tracking recent reads will allow users to manage their book collection and stay on top of their reading goals.

Click a link to view or update the database

- [Books](#)
- [Authors](#)
- [Genres](#)
- [Reviews](#)
- [Users](#)

Books Page (Create, Read, Update, Delete)

Users can read the database's list of books, as well as add new books to the masterlist. The add function can be found directly underneath the Books table. When a user clicks on "Delete", they can delete a book from the list. Additionally, users can update a book's isbn or publication date by filling out the Update a Book form at the bottom of the page.

Books

book_id	title	isbn	publication_date	delete
1	A Tale for the Time Being	9780143124870	Tue Dec 31 2013 00:00:00 GMT-0800 (Pacific Standard Time)	Delete
2	The Hobbit	9780547928227	Tue Sep 18 2012 00:00:00 GMT-0700 (Pacific Daylight Time)	Delete
3	The Way of Kings	9780765376671	Tue Mar 04 2014 00:00:00 GMT-0800 (Pacific Standard Time)	Delete
4	Villette (Penguin Classics)	9780140434798	Tue Dec 28 2004 00:00:00 GMT-0800 (Pacific Standard Time)	Delete

Add a new book

To add a new book, please enter the following information below:

Title: ISBN: Publication Date:

Update a Book

To update a book's isbn or publication date, select desired title from the dropdown

Title: ISBN: Publication Date:

Users Page

Users (Create, Read, Delete)

This page displays a list of users with the username, email, password, and reading goal. A user can be created by filling out the Add new user form, and users can be deleted by pressing the Delete button to the right of the user.

Users

user_id	username	email	password	reading_goal	delete
1	fake_user123	fake_user@gmail.com	1234!!	30	<button>Delete</button>
2	second_user	Second@gmail.com	password	10	<button>Delete</button>
3	third_user	third@gmail.com	anotherpassword	15	<button>Delete</button>
4	fourth_user	fourth@gmail.com	lastpassword	45	<button>Delete</button>

Add new user

To add a new user, please enter the following information below:

Username: Email: Password: Reading Goal: Submit

Users_read_books (Create, Read, Delete)

Intersection table between Books and Users. Read shows the relationship between the username and the book that they read with the read date. The relationship can be deleted, and new relationships can be added by submitting the Add a new read relationship at the bottom.

Users_read_books

Username	Book Title	Read Date	Delete
fake_user123	A Tale for the Time Being	Wed Sep 25 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>
second_user	The Hobbit	Thu Oct 31 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>
third_user	The Way of Kings	Wed Jan 15 2020 00:00:00 GMT-0800 (Pacific Standard Time)	<button>Delete</button>
fourth_user	Villette (Penguin Classics)	Sat Jun 08 2019 00:00:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>

Add a new read relationship

User: Book Title: Read Date:

Reviews Page (Create, Read, Update, Delete)

This page displays the reviews made by users with the book that they are reviewing. New reviews can be added by filling out the Add a new review underneath the table. Drop-down functions allow the user to choose username and book title rather than having to type it in. Reviews can also be deleted by clicking on the Delete button to the right of the review. There is also an update function for reviews, which can be accessed by clicking on the Edit button next to the review that needs to be updated. The update is for a M:N relationship. The delete is also for a M:N relationship.

Reviews

Review ID	Review Date	Review Score	Review Text	User	Book	Delete	Edit
1	Wed Sep 25 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	5	Great book	fake_user123	A Tale for the Time Being	Delete	Edit
2	Thu Oct 31 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	5	Loved this one	second_user	The Hobbit	Delete	Edit
3	Wed Jan 15 2020 00:00:00 GMT-0800 (Pacific Standard Time)	5	My favorite!	third_user	The Way of Kings	Delete	Edit
4	Sat Jun 08 2019 00:00:00 GMT-0700 (Pacific Daylight Time)	5	New favorite	fourth_user	Villette (Penguin Classics)	Delete	Edit

Add a new review

To add a new review, please enter the following information below:

Date: Score: Review Text: User: Book Title:

Update a Review

To update a Review, fill out this form

Date: Score: Review Text: User: Book Title:

When updating a review, user can be made nullable:

Reviews

Review ID	Review Date	Review Score	Review Text	User	Book	Delete	Edit
1	Wed Sep 25 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	5	Great book	fake_user123	A Tale for the Time Being	Delete	Edit
2	Thu Oct 31 2024 00:00:00 GMT-0700 (Pacific Daylight Time)	5	Loved this one	second_user	The Hobbit	Delete	Edit
3	Wed Jan 15 2020 00:00:00 GMT-0800 (Pacific Standard Time)	5	My favorite!	third_user	The Way of Kings	Delete	Edit
4	Sat Jun 08 2019 00:00:00 GMT-0700 (Pacific Daylight Time)	5	New favorite	fourth_user	Villette (Penguin Classics)	Delete	Edit

Add a new review

To add a new review, please enter the following information below:

Date:

Score:

Review Text:

User:

Select a User

Make null

fake_user123

fourth_user

second_user

third_user

Book Title:

Select a Book Title

Add Review

To update a Review, fill out this form

Date:

Score:

Review Text:

User:

fake_user123

Book Title:

A Tale for the Time Being

update Review

Authors Page

Authors (Create, Read, Delete)

Displays list of authors. A new author can be created by filling out the Add a new author form underneath the table. Authors can additionally be deleted by clicking on the Delete button next to the right of the desired author.

Authors

author_id	first_name	last_name	date_of_birth	bio	delete
1	Ruth	Ozeki	Mon Mar 12 1956 00:00:00 GMT-0800 (Pacific Daylight Time)	American-Canadian author, filmmaker and Zen Buddhist priest	Delete
2	John	Tolkein	Sun Jan 03 1892 00:00:00 GMT-0800 (Pacific Standard Time)	English writer and philologist	Delete
3	Brandon	Sanderson	Fri Dec 19 1975 00:00:00 GMT-0800 (Pacific Standard Time)	American, New York Times best-selling author of high fantasy, science fiction, and young adult books	Delete
4	Charlotte	Bronte	Sun Apr 21 1816 00:00:00 GMT-0752 (Pacific Daylight Time)	English novelist and poet, the eldest of the three Brontë sisters	Delete

Add a new author

To add a new author, please enter the following information below:

First Name:

Last Name:

Date of Birth:

Bio:

Enter a brief biography here:

Submit

Books_has_authors (Create, Read, Delete)

Displays the relationship between Books and Authors. To add a new relationship, the Add a new write relationship can be filled out underneath the table. Relationships can also be deleted by clicking on the Delete button to the right.

Books_has_authors

Book Title	Author Name	Delete
A Tale for the Time Being	Ruth Ozeki	<button>Delete</button>
The Hobbit	John Tolkein	<button>Delete</button>
The Way of Kings	Brandon Sanderson	<button>Delete</button>
Villette (Penguin Classics)	Charlotte Bronte	<button>Delete</button>

Add a new write relationship

Book Title: Book Title:

Genres Page

Genres (Create, Read, Delete)

Displays list of genres. A new genre can be added via the Add a new genre form underneath the table. Genres can be deleted by clicking on the Delete button to the right.

Genres

genre_id	name	description	delete
1	Fantasy	Includes fantastical elements, such as magic or supernatural forces.	<button>Delete</button>
2	Mystery	Plot revolves around a crime or mystery to be solved.	<button>Delete</button>
3	Literary Fiction	Character-driven and introspective.	<button>Delete</button>
4	Non-Fiction	Events, people, and concepts based on fact and reality.	<button>Delete</button>

Add a new genre

To add a new genre, please enter the following information below:

Name: Description:

Books_has_genres (Create, Read, Delete)

Displays the relationship between Books and Genres. To add a new relationship, the Relate a book to a genre form can be filled out underneath the table. Relationships can also be deleted by clicking on the Delete button to the right.

Books_has_genres

Book Title	Genre	Delete
A Tale for the Time Being	Literary Fiction	<button>Delete</button>
The Hobbit	Fantasy	<button>Delete</button>
The Way of Kings	Fantasy	<button>Delete</button>
Villette (Penguin Classics)	Literary Fiction	<button>Delete</button>

Relate a book to a genre

Book Title: Genre:

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